AIDCF

AI Data Centre Facilities Assurance Framework

END-TO-END TRACEABILITY AND MACHINE-EXECUTABLE DATA STANDARD

A canonical information contract for identifiers, objects, relationships, lineage, vocabularies, validation, exchange, change impact and conformance

Document control

Field	Value
Public ID	AIDCF-014
Short code	TDS (requirement IDs TDS-REQ-NNN)
Title	End-to-End Traceability and Machine-Executable Data Standard
Version	1.1
Date	2026-10-10
Status	Draft for controlled review
Author	Siva Sethumadhavan
Depends on	AIDCF-002, AIDCF-003, AIDCF-004, AIDCF-005, AIDCF-006, AIDCF-007A, AIDCF-008, AIDCF-009, AIDCF-010, AIDCF-011, AIDCF-012, AIDCF-013
Normative companion	AIDCF-016 Machine-Executable Data Model Standard (serialisation, schema and ontology package)
Used by	AIDCF-015 (reference implementation and synthetic dataset), AIDCF-017, AIDCF-018, AIDCF-019, AIDCF-VR-001
Boundary	Technical conformance to this standard is not engineering approval, assurance or certification.

Revision history

Version	Date	Summary
1.0	Not stated	Initial draft (internal ID AIDCF-TDS-001): object, relationship and registry inventories, serialisation list, business rules, conformance tests and templates.
1.1	2026-10-10	Public ID AIDCF-014 adopted; vocabularies aligned to the canonical specification (control result, gate result, decision state and lifecycle, operating state) with legacy alias tables; namespace and version IRI declared; AIDCF-016 named as normative serialisation companion; 48 classes and 41 relationship types fully defined; release manifest requirements specified; boilerplate rows replaced; numbered requirements TDS-REQ-001 onward; §15–§17 aligned to published AIDCF-016 and registry catalogue.

Executive thesis

Every material assurance conclusion shall be reconstructable, by a person or a machine, from stable identities, pinned source versions, typed relationships, evidence and recorded human review. Missing data is UNKNOWN, conflicting data is a BaselineIssue or a CHALLENGES edge, and neither is silently resolved.

AIDCF-014 is the information contract of the framework. It defines *what* must be recorded and linked (classes, relationships, vocabularies, lineage, business rules and conformance tests). AIDCF-016 defines *how* it is serialised (JSON Schema, JSON-LD context, OWL ontology, SHACL shapes, CSV registry layouts and the versioned schema and ontology package). Where the two differ, AIDCF-014 governs meaning and AIDCF-016 governs syntax; a difference in meaning is a BaselineIssue (§6).

Canonical assurance chain

The chain below is the only assurance chain in AIDCF and is based on the ISO/IEC/IEEE 15026-2 assurance-case structure. Each step is a class in §7; each arrow is one or more relationship types in §9; each traceability rule in §11 tests one or more arrows.

Step	Chain element	Principal classes (§7)	Forward relationship(s) (§9)
1	Decision Question	Assessment (decisionQuestion), AssuranceProgram	GOVERNED_BY (Assessment to AssuranceProgram)
2	Capability Instance	CapabilityInstance	HAS_OUTCOME
3	Service Outcome	ServiceOutcome	REQUIRES_FUNCTION
4	Required Function	RequiredFunction	IMPLEMENTED_BY
5	Architecture	ArchitectureSnapshot, InfrastructureSystem, AssetInstance, Interface, Resource, CapacityAssertion	COMPOSED_OF, CONNECTS_TO, CONSUMES_RESOURCE, CONTROLLED_BY, MONITORED_BY, LOCATED_IN
6	Dependency and Fault Domain	Dependency, FaultDomain	DEPENDS_ON, SHARES_FAULT_DOMAIN_WITH

Step	Chain element	Principal classes (§7)	Forward relationship(s) (§9)
7	Threat / Hazard / Failure Scenario	Threat, Hazard, FailureMode, FailureScenario	EXPOSED_TO, INITIATES, PROPAGATES_TO
8	Safeguard	Safeguard	PROTECTED_BY, RECOVERS_THROUGH
9	Control	ControlDomain, ControlFamily, Control	GOVERNED_BY (Safeguard to Control)
10	Evidence Requirement	EvidenceRequirement	HAS_EVIDENCE_REQUIREMENT
11	Evidence Object	EvidenceObject, EvidencePackage	EVIDENCED_BY, SUPPORTS, CHALLENGES, QUALIFIES
12	Control Assessment (result)	Assessment, AssessmentProcedure, Sample, TestRecord, ControlAssessment	ASSESSED_BY, RESULTS_IN
13	Finding / UNKNOWN Record	Finding, UnknownRecord	HAS_FINDING, HAS_UNKNOWN
14	Gate Result	CriticalGate, GateResult	EVALUATES_GATE, CONSTRAINS
15	Score Snapshot and Decision Recommendation	ScoreSnapshot, DecisionRecommendation	SUMMARISED_BY, RECOMMENDS
16	Assurance Decision	AssuranceDecision, ReviewRecord	REVIEWED_BY, APPROVED_BY
Loop	Report and Reassessment Trigger	AssuranceStatement, AssuranceReport, Action, ReassessmentTrigger, Event	REPORTED_IN, REQUIRES_ACTION, CLOSED_BY, INVALIDATED_BY, TRIGGERS_REASSESSMENT

Assurance-case mapping: the top claim and sub-claims at steps 3 to 8 are Claims; decomposition by function, dependency and scenario is the Argument/Strategy; steps 10 to 12 are Evidence; CHALLENGES edges, Findings and UNKNOWN Records are defeaters; operating states, boundary and exclusions are Context.

1 Authority and precedence

TDS-REQ-001 Each canonical class, relationship type and vocabulary shall have exactly one semantic owner artifact (§7, §9, §11A). A schema, ontology, shape or tool shall not redefine a term whose semantic owner is another artifact.

TDS-REQ-002 A schema, shape, profile, extension or tool shall not weaken a normative rule of AIDCF-006 to AIDCF-012 (for example by making a mandatory link optional, accepting an undeclared state code or permitting a score to override a gate).

TDS-REQ-003 Where two sources disagree on a count, identifier, title, definition, mapping or status, implementations shall not choose one silently; they shall create a BaselineIssue (§6).

TDS-REQ-004 Every exchange or publication of AIDCF data shall be bound to a ReleaseManifest (§4) that pins the versions and hashes of the artifacts, registries, vocabularies, schemas and ontology it relies on.

TDS-REQ-005 This standard is technology neutral. Conformance shall be evaluated on the information content and behaviour defined here, not on any particular database, graph store, programming language or product.

TDS-REQ-006 Automated components may validate, derive, propose and flag. They shall not approve a material fact, a control result, a gate result, a decision or a report. Approval is recorded only through a ReviewRecord linked by REVIEWED_BY or APPROVED_BY to an accountable person.

TDS-REQ-007 Records shall not be destructively overwritten. Change is expressed by a new version or by a superseding record (SUPERSEDES), and prior versions shall remain retrievable.

TDS-REQ-008 Structural validity (schema, shape or test pass) shall not be presented as evidence that a claim is true, that a design is adequate, or that any assurance level has been achieved.

Precedence when sources conflict: (1) AIDCF-007A amendment instruments and the canonical specification baseline for identifiers and cardinality; (2) the semantic owner artifact named in §7 or §9 for meaning; (3) this standard for structure, linkage and validation; (4) AIDCF-016 for serialisation syntax; (5) implementation documentation. A conflict at any level is recorded as a BaselineIssue.

2 Constitutional invariants

The invariants below are properties that every conformant dataset shall exhibit at all times. Each is testable by the conformance tests in §31.

Requirement	Invariant	Testable statement
TDS-REQ-009	Capability and outcomes are primary	Every Assessment references exactly one CapabilityInstance and at least one ServiceOutcome; no ControlAssessment exists without a scope that resolves to a CapabilityInstance.
TDS-REQ-010	Assertions require evidence or UNKNOWN	Every material assertion (CapacityAssertion, Dependency, FaultDomain independence, ControlAssessment, GateResult) has at least one EVIDENCED_BY edge to an EvidenceObject or a HAS_UNKNOWN edge.
TDS-REQ-011	UNKNOWN is first-class	An unresolved fact is an UnknownRecord with type (U1–U10) and severity (US1–US5); a null value without nullReason is invalid.
TDS-REQ-012	Gates are separate from scores	ScoreSnapshot never sets or modifies a GateResult; an AssuranceDecision references GateResults directly, not through scores.

Requirement	Invariant	Testable statement
TDS-REQ-013	Claim types are distinct	A ControlAssessment records exactly one claim type (DESIGN, IMPLEMENTATION, OPERATION, EFFECTIVENESS, RESILIENCE); a result for one claim type is not reused for another.
TDS-REQ-014	Dependencies are visible	Every DEPENDS_ON edge references a Dependency record with direction, conditions (operating states) and criticality, or an UnknownRecord for each missing attribute.
TDS-REQ-015	Lineage is mandatory	Every derived record (ControlAssessment, GateResult, ScoreSnapshot, DecisionRecommendation, AssuranceDecision, AssuranceStatement) can be traced to its inputs by the rules in §11.
TDS-REQ-016	Change can invalidate	An Event that matches a ReassessmentTrigger can create INVALIDATED_BY edges and change dependent decision lifecycle to SUSPENDED after human disposition.
TDS-REQ-017	Automation proposes	Records created by automated components carry reviewState CANDIDATE until a ReviewRecord accepts them.
TDS-REQ-018	AIDCF and AITG conclusions are separate	An AIDCF AssuranceDecision is never asserted as, or derived automatically from, an AI Trust Graph conclusion, or vice versa; the only cross-framework link is INTEGRATION_HANDOFF (AIDCF-005).
TDS-REQ-019	Connectivity is not causality	CONNECTS_TO does not imply DEPENDS_ON, PROPAGATES_TO, CONTROLLED_BY or reachability; each must be asserted and evidenced separately.
TDS-REQ-020	Sensitive paths are controlled	Records and edges describing exploitable paths (for example remote-access routes to OT controllers) carry a classification and a DistributionControl, and are excluded from exports that do not authorise them.

3 Conformance levels

Conformance levels are cumulative: level T*n* requires all criteria of T0 to T*n*−1. An implementation claims the highest level whose criteria it meets in full, stating the ReleaseManifest it was tested against.

Level	Name	Criteria (all mandatory for the level)
T0	Vocabulary	Uses canonical codes of §11A and the canonical specification; legacy codes accepted only on input and mapped through the alias tables; no undeclared code is emitted.
T1	Records and identifiers	All records use the common envelope (§8) and the identifier rules (§5); IDs are unique and stable across versions.
T2	Traceability	All traceability rules TR-01 to TR-18 (§11) are evaluated; every gap is reported as a validation failure or an UnknownRecord.
T3	Structural validation	All records validate against the AIDCF-016 JSON Schema package for the pinned version, with additionalProperties false.
T4	Graph validation	The graph projection validates against the AIDCF-016 SHACL shapes, including the domain, range and cardinality constraints of §9.
T5	Decision-ready	Business rules BR-001 to BR-015 (§18) are enforced; gate roll-up and decision ceilings are reproduced from inputs; reports are checked against decisions.
T6	Continuing assurance	Events (§20, §21) are ingested; the change-impact contract (§22) is executed; affected records are flagged and routed for human disposition.
T7	Test-suite conformance	The full conformance test suite (§31) for the pinned release passes, and the results are published as a ValidationResult set referenced by the ReleaseManifest.

TDS-REQ-021 A conformance claim shall state the level, the ReleaseManifest identifier, the test-suite version and the date of testing. A claim of T5 or above shall be supported by a published ValidationResult set.

4 Release manifest

A ReleaseManifest is the signed, immutable bill of materials for one AIDCF release. All other records are interpreted against it.

TDS-REQ-022 Every AIDCF release shall publish exactly one ReleaseManifest, serialised as defined in AIDCF-016, containing the fields below.

Field	Content and rule
releaseId	Stable identifier of the release (pattern AIDCF-REL-YYYY.MM.N); never reused.
releaseStatus	Lifecycle code (§11A.1). A release containing any artifact whose status is "Draft for controlled review" shall itself be no higher than REVIEWED.
artifactManifest	One entry per artifact: public ID (AIDCF-NNN, AIDCF-007A, AIDCF-VR-001, AIDCF-WP-001), title, version, status, repository path, media type and SHA-256 digest of the exact file.
registryVersions	One entry per registry (§17): name, version, record count and digest. The control registry entry shall assert the cardinality 15 domains, 90 families, 360 controls and shall fail validation if the counts differ.
vocabularyVersions	Version and digest of each controlled vocabulary file (registries/vocabularies/*.csv).
schemaVersions	Version and digest of each JSON Schema, the JSON-LD context, the OWL ontology and the SHACL shapes in the AIDCF-016 package, including the ontology version IRI.
approvedCardinality	The cardinality decision record in force (AIDCF-007A-AMD-001) and any later amendment that applies.
knownIssues	References to every open BaselineIssue, with status BLOCKED or CONDITIONALLY_BASELINED.
migrationSet	References to MigrationRecords that convert records from the previous release, including vocabulary alias migrations.

Field	Content and rule
testSuite	Conformance test-suite version, digest, and references to the ValidationResult set produced against this release.
approvals	ReviewRecord references for each required review (§38), each with outcome APPROVED, PENDING or REJECTED. PENDING approvals shall be listed, not omitted.
classification	Classification code (§11A.8) of the manifest itself.
supersedes	releaseId of the previous release, if any.
integrity	Digest algorithm, canonicalisation method and detached signature over the manifest.

TDS-REQ-023 Digests shall be SHA-256 over the exact bytes of each file. The manifest itself shall be canonicalised with the JSON Canonicalization Scheme (IETF RFC 8785) before its digest is computed and signed.

TDS-REQ-024 A published ReleaseManifest shall not be modified. A correction is a new ReleaseManifest that SUPERSEDES the previous one.

TDS-REQ-025 A consumer shall reject, or treat as UNKNOWN, any record whose referenced artifact, registry, vocabulary or schema version is not listed in the ReleaseManifest the record declares.

TDS-REQ-026 A release whose approvals include any PENDING entry shall be labelled "Draft for controlled review" in every human-readable rendering.

5 Identifier standard

5.1 Namespace

The AIDCF ontology namespace is `https://w3id.org/aidcf/ontology#` with the prefix `aidcf:`. The version IRI of the ontology aligned to this release is `https://w3id.org/aidcf/ontology/1.3`. Registration of the `w3id.org/aidcf` redirect is pending; until it is confirmed, these IRIs are identifiers only and shall not be assumed to resolve. AIDCF-016 defines the instance-data IRI pattern and the JSON-LD context that maps JSON keys to these IRIs.

5.2 Rules

TDS-REQ-027 Every record shall have an identifier that is unique within its class across all releases and is never reused for a different meaning.

TDS-REQ-028 A change of meaning shall create a new identifier and a SUPERSEDES edge from the new record to the old. A change that does not alter meaning (correction of a typographical error, a disambiguating title change recorded by amendment) shall create a new version of the same identifier.

TDS-REQ-029 Version shall be carried separately from identity (the `version` envelope field); identifiers shall not embed version numbers, except release identifiers and ontology version IRIs.

TDS-REQ-030 Controlled codes shall be used exactly as defined in the canonical patterns below; case is significant.

Object	Pattern	Example	Semantic owner
Artifact	AIDCF-NNN, AIDCF-007A, AIDCF-018-DOM, AIDCF-VR-NNN, AIDCF-WP-NNN	AIDCF-008	AIDCF-014
Control domain	three-letter code	THM	AIDCF-007A
Control family	DOM-NN (01–06)	THM-03	AIDCF-007A
Control	MCF-DOM-NNN (001–024)	MCF-THM-010	AIDCF-007A
Critical gate	CG-NN (01–15)	CG-06	AIDCF-010
Amendment	AIDCF-007A-AMD-NNN	AIDCF-007A-AMD-002	AIDCF-007A
Requirement	SHORT-REQ-NNN	TDS-REQ-001	Owning artifact
Release	AIDCF-REL-YYYY.MM.N	AIDCF-REL-2026.10.1	AIDCF-014

TDS-REQ-031 External identifiers (asset tags, CMDB keys, ticket numbers, vendor serial numbers, AITG identifiers) shall be recorded in `externalIdentifiers` as typed pairs (scheme, value) and shall never be used as the AIDCF identifier.

TDS-REQ-032 Two identifiers that denote the same real-world entity shall be linked only by a reviewed shared-identity assertion carrying evidence and a ReviewRecord; automated matching produces a candidate only.

TDS-REQ-033 Identifiers minted by automated discovery shall be marked CANDIDATE and shall not be referenced by an approved decision until accepted by review.

TDS-REQ-034 An identifier collision (same identifier, different meaning) shall block release and create a BaselineIssue with status BLOCKED.

6 Baseline harmonisation

TDS-REQ-035 Conflicts between AIDCF sources shall be recorded as BaselineIssue records, never resolved in place.

Attribute	Rule
Trigger	Conflicting counts, identifiers, titles, definitions, mappings, vocabulary codes or statuses between two or more pinned sources.
Sources	Each conflicting source is referenced with artifact ID, version, location and the exact conflicting value; original values are preserved verbatim.
Status	BLOCKED (release cannot proceed; dependent records are UNKNOWN) or CONDITIONALLY_BASELINED (a provisional value is adopted with a stated condition and expiry).
Resolution	An approved decision (ReviewRecord), the registry update, a MigrationRecord and, where meaning changes, a SUPERSEDES edge.
Closure	CLOSED_BY the ReviewRecord and MigrationRecord; the issue remains in history.

Example: the cardinality conflict between the AIDCF-007 v1.0 workbook (15/90/360) and narrative statements of 14/84/336 is a resolved BaselineIssue, closed by AIDCF-007A-AMD-001.

7 Canonical object registry

TDS-REQ-036 The 48 classes below are the canonical object classes of AIDCF. Each is an `aidcf:` class in the AIDCF-016 ontology; no implementation shall add a class to this list except by extension (§33).

The owner artifact is the semantic owner of the class's meaning (§1). The chain step refers to the canonical assurance chain.

#	Class	Definition	Owner artifact	Chain step
1	ReleaseManifest	Signed, immutable bill of materials pinning every artifact, registry, vocabulary, schema and ontology version of one release.	AIDCF-014	Context
2	BaselineIssue	Recorded conflict between pinned sources, with original values, status and resolution lineage.	AIDCF-014	Context
3	CapabilityInstance	A specific deployment of an AI capability bounded by ownership, location, architecture, capacity, criticality and obligations.	AIDCF-003	2
4	ServiceOutcome	A measurable or assessable result the CapabilityInstance must achieve, with tolerance, duration and operating-state conditions.	AIDCF-003	3
5	RequiredFunction	A function whose loss or degradation materially affects one or more ServiceOutcomes (for example "remove 100% of rack heat at design inlet temperature").	AIDCF-003	4
6	ArchitectureSnapshot	A versioned, approved and immutable view of the architecture in scope at a point in time, against which assessments are performed.	AIDCF-013	5
7	InfrastructureSystem	A coordinated set of assets and interfaces providing one or more functions (for example a CDU secondary loop, a UPS system, a scale-out fabric).	AIDCF-003	5
8	AssetInstance	An identifiable real-world item, service, procedure or resource provider represented where material (for example a switchboard, a CDU, a GPU node).	AIDCF-003	5
9	Interface	A defined point or mechanism of exchange between entities (electrical, thermal, fluid, network, management, data, physical, operational, provider).	AIDCF-003	5
10	Resource	A consumable or constrained input required for operation (electrical capacity, cooling capacity, water, fuel, rack space, accelerator hours).	AIDCF-003	5
11	Dependency	A reified reliance of one entity on another entity, resource, condition or decision, with direction, strength, states and substitutability.	AIDCF-003	6
12	FaultDomain	A set of entities expected to be affected together by a common cause of failure, with its boundary and blast radius.	AIDCF-003	6
13	OperatingState	A defined condition of operation (NORMAL, DEGRADED, MAINTENANCE, EMERGENCY, RECOVERY, TRANSITION, SHUTDOWN, FAILED, UNKNOWN).	AIDCF-003	Context
14	CapacityAssertion	A conditioned claim about a capacity value: metric, unit, subject, operating state, environment, duration, derating, evidence date and confidence.	AIDCF-003	5
15	Threat	An actor-driven or intentional source of potential harm (for example unauthorised remote command of a cooling controller).	AIDCF-004	7
16	Hazard	A non-intentional source of potential harm (for example utility voltage sag, coolant leak, heatwave, battery thermal runaway).	AIDCF-004	7
17	FailureMode	The manner in which an entity fails to perform its function (for example pump seizure, breaker failure to trip, optic degradation).	AIDCF-004	7
18	FailureScenario	An ordered path from initiating source through propagation steps and safeguards to residual state and outcome effect, from a stated pre-state.	AIDCF-004	7
19	Safeguard	A real-world measure that prevents, detects, contains, corrects or recovers from a failure, threat or hazard.	AIDCF-004	8
20	ControlDomain	One of the 15 AIDCF-007A control domains.	AIDCF-007A	9
21	ControlFamily	One of the 90 AIDCF-007A control families (six per domain).	AIDCF-007A	9
22	Control	One of the 360 AIDCF-007A normative controls: objective, requirement, criteria, evidence profile, procedure and gate mappings.	AIDCF-007A	9

#	Class	Definition	Owner artifact	Chain step
23	EvidenceRequirement	The evidence a control or gate needs for a claim type: classes, quality and sufficiency floors, state and period coverage, independence.	AIDCF-008	10
24	EvidenceObject	A record or observation, with provenance and integrity, used to support, challenge or qualify an assertion.	AIDCF-008	11
25	EvidencePackage	A sealed, manifest-listed set of EvidenceObjects collected for an assessment or gate, with custody and integrity metadata.	AIDCF-008	11
26	AssuranceProgram	The programme within which assessments are planned and decisions issued: assurance level target, profile, cadence and authority.	AIDCF-011	1
27	Assessment	One bounded assessment answering one decision question for one CapabilityInstance, scope and ArchitectureSnapshot.	AIDCF-009	1, 12
28	AssessmentProcedure	The ordered method steps used to assess one control for one claim type, using canonical assessment methods.	AIDCF-009	12
29	Sample	A defined selection of population items examined by a procedure, with population, selection method and size rationale.	AIDCF-009	12
30	TestRecord	The record of an authorised test or scenario exercise: authority, preconditions, safety controls, steps, observations and outcome.	AIDCF-009	12
31	ControlAssessment	The result of assessing one control, for one claim type, in one scope: result code, confidence, evidence, method, assessor and lifecycle.	AIDCF-009	12
32	Finding	A documented deficiency against a stated criterion, with condition, cause, consequence, evidence and gate and score effect.	AIDCF-009	13
33	UnknownRecord	A material fact that cannot be determined, with type (U1–U10), severity (US1–US5), reason, decision effect, evidence needed and owner.	AIDCF-008	13
34	CriticalGate	One of the 15 non-compensating critical gates CG-01 to CG-15, with scope, criteria and evidence floor.	AIDCF-010	14
35	GateResult	The evaluated status of one CriticalGate in one assessment scope (PASS, FAIL, UNKNOWN, APPROVED_NA, NOT_ASSESSED) with its inputs.	AIDCF-010	14
36	ScoreSnapshot	An immutable, reproducible computation of control, family and domain scores, coverage and band from pinned inputs and profile.	AIDCF-010	15
37	DecisionRecommendation	The decision state, conditions, ceilings and limitations proposed from gate results, UNKNOWNs and scores, before authority review.	AIDCF-010	15
38	ReviewRecord	A recorded human review or approval: reviewer, authority, independence, scope reviewed, outcome and date.	AIDCF-009	16
39	AssuranceDecision	The decision of the accountable authority: decision state, conditions, scope, validity period and lifecycle.	AIDCF-010	16
40	AssuranceStatement	The bounded, disclosed wording of an AssuranceDecision for a named audience, which cannot exceed the decision.	AIDCF-012	Loop
41	AssuranceReport	The issued report carrying statements, scope, findings, UNKNOWNs, limitations, distribution and validity.	AIDCF-012	Loop
42	Action	A remediation, evidence-collection or condition-fulfilment task with owner, due date and closure criteria.	AIDCF-011	Loop
43	ReassessmentTrigger	A declared rule describing which events or conditions require reassessment of which records.	AIDCF-011	Loop
44	Event	An observed or reported occurrence (change, incident, test failure, evidence expiry) ingested for impact analysis.	AIDCF-014	Loop
45	IntegrationHandoff	A reviewed exchange of identity or findings between AIDCF and the AI Trust Graph, preserving each framework's ownership.	AIDCF-005	Context
46	ValidationResult	The outcome of a structural, semantic or business-rule validation of a record or package against a pinned release.	AIDCF-014	Context
47	MigrationRecord	The record of converting data between versions or vocabularies: source, target, rule, mapping, exceptions and validation.	AIDCF-014	Context
48	DistributionControl	The classification-bound disclosure rule for a record, edge, package or report: permitted audiences, purpose, redaction and expiry.	AIDCF-014	Context

Reference classes. Relationship ranges in §9 also use three AIDCF-003 classes that are not among the 48 canonical assurance classes: **Location** (site, building, hall, room, row, rack position or logical placement), **PartyEntity** (organisation, role, actor or team) and **Component** (a constituent part of an asset represented where material). They are imported from the AIDCF-003 module of the ontology and are referenced, not redefined, here.

8 Common object envelope

TDS-REQ-037 Every record of every class in §7 shall carry the envelope below. A field marked "Mandatory" shall be present and non-empty; a field marked "Conditional" shall be present when its condition holds, and otherwise may be omitted. A mandatory value that cannot be supplied shall be null with a `nullReason` (§13).

Field	Type	Obligation	Meaning
id	identifier (§5)	Mandatory	Stable identity of the record.
type	class name (§7)	Mandatory	Discriminator; exactly one canonical class or registered extension class.
semanticOwner	artifact public ID	Mandatory	Artifact that owns the meaning of this record's class.
version	semantic version string	Mandatory	Version of this record (§27).

Field	Type	Obligation	Meaning
status	lifecycle code (§11A.1)	Mandatory	Record lifecycle.
releaseRef	releaseId	Mandatory	ReleaseManifest against which the record is valid.
sourceArtifact	artifact public ID	Mandatory for registry-derived records	Artifact the record was derived from.
sourceRecord	typed reference	Conditional: derived or imported	Originating record in a source system (CMDB, EPMS historian, ticketing).
name	string	Mandatory	Human-readable name.
description	string	Conditional: class profile requires it	Definition or narrative.
scope	scope object	Mandatory for assessable classes	CapabilityInstance, boundary and ArchitectureSnapshot the record applies to.
effectiveFrom	ISO 8601 date-time	Mandatory	Start of validity.
effectiveTo	ISO 8601 date-time	Conditional: time-bounded records	End of validity (evidence expiry, decision validity).
createdAt, createdBy	date-time, party reference	Mandatory	Creation provenance.
modifiedAt, modifiedBy	date-time, party reference	Mandatory once modified	Last modification provenance.
classification	classification code (§11A.8)	Mandatory	Sensitivity of the record.
distributionControlRef	DistributionControl reference	Conditional: classification above INTERNAL	Disclosure rule.
provenance	provenance object (§26)	Mandatory	Source system, method, transformation version, hash.
reviewState	assertion code (§11A.2)	Mandatory for assertion-bearing classes	CANDIDATE until reviewed.
supersedes	list of identifiers	Conditional: replaces earlier record	Records this one supersedes.
externalIdentifiers	list of (scheme, value)	Optional	Typed external keys (§5).
extensions	object keyed by registered namespace	Optional	Extension properties (§33).

8A Object profiles

TDS-REQ-038 In addition to the envelope, records of the classes below shall carry the profile fields listed. Each field shall hold a value, an approved NOT_APPLICABLE with rationale, or a reference to an UnknownRecord; an empty field is invalid.

Capability profile (CapabilityInstance)

Field	Content
owner	PartyEntity accountable for the capability.
lifecycle	Lifecycle stage (PLAN, DESIGN, BUILD, COMMISSION, OPERATE, CHANGE, DECOMMISSION).
workload	Workload classes served (training, fine-tuning, inference, data processing) and their characteristics.
architectureBaseline	Current approved ArchitectureSnapshot.
outcomeIds	ServiceOutcomes (via HAS_OUTCOME).
criticality	Criticality classification per AIDCF-007A CAP-03.
boundary	Inclusions, exclusions and external providers.
demandProfile	Baseline, peak and transient demand, growth forecast (AIDCF-007A CAP-04).
operatingEnvelope	Load, environmental, duration and state-specific limits (AIDCF-007A CAP-05).

Architecture profile (ArchitectureSnapshot)

Field	Content
capabilityId	CapabilityInstance the snapshot describes.
systems, assets, interfaces, resources	Member records (via COMPOSED_OF).
dependencies, faultDomains	Dependency and FaultDomain records in scope.
states	OperatingStates represented and their transition rules.
assertions	CapacityAssertions and independence assertions.
asBuiltReconciliation	Reference to the reconciliation evidence between design and observed state.
validation	ValidationResults for the snapshot.

Dependency profile

Field	Content
sourceId, targetId	Consumer and provider entities.
dependencyType	ELECTRICAL, THERMAL, FLUID, NETWORK, CONTROL, TIMING, IDENTITY, CONFIGURATION, DATA, HUMAN, SUPPLIER, UTILITY, RECOVERY.
strength	ABSOLUTE, MATERIAL, CONDITIONAL, SUPPORTING, RECOVERY_ONLY, UNKNOWN (AIDCF-003).
direction	Effect direction from provider to consumer.
conditions	OperatingStates in which the reliance is material.
criticality	Effect on ServiceOutcome if lost.
substitutability	Alternatives and time to substitute.
sharedMode	FaultDomain reference where the provider is shared.
evidenceRefs, confidence	Evidence and confidence (CONF-0 to CONF-4).

FailureScenario profile

Field	Content
preState	OperatingState and configuration before initiation.
initiatingSource	Threat, Hazard or FailureMode (via INITIATES).
steps	Ordered PROPAGATES_TO edges with timing.
safeguards	Safeguards expected to act at each step (via PROTECTED_BY).
residualState	State after safeguards act or fail.
outcomeEffects	Effect on each ServiceOutcome against tolerance.
evidenceRefs, confidence	Analysis or test evidence and confidence.

Safeguard profile

Field	Content
function	PREVENT, DETECT, CONTAIN, CORRECT or RECOVER.
protectedSubject	Entity or outcome protected (inverse of PROTECTED_BY).
activation	Automatic, manual or hybrid; trigger and time to act.
authority	PartyEntity or control system authorised to activate.
dependencies	What the safeguard itself depends on (power, control, network, people).
stateCoverage	OperatingStates in which it is effective.
failureModes	How the safeguard itself can fail.
controlRefs	Controls that govern it (GOVERNED_BY).

Control profile

Field	Content
domainId, familyId	Canonical codes.
objective, requirement, assessmentCriteria	As in the AIDCF-007A registry.
applicability	When the control applies and does not.
assessmentMethods	Canonical method codes.
claimType, requiredAssuranceLevel	Primary claim type and AL2–AL5.
evidenceProfile	EvidenceRequirement reference (HAS_EVIDENCE_REQUIREMENT).
gateMappings	List of (gateId, role) with role CRITICAL or SUPPORTING (EVALUATES_GATE).

Evidence profile (EvidenceObject)

Field	Content
evidenceClass	E1 to E7.
source, collector, custodian	Provenance and custody.
assertionRefs	Assertions supported, challenged or qualified.

Field	Content
period, state	Observation period and OperatingStates covered.
integrity	Integrity level I0 to I5 and digest.
qualityVector	P, F, C, R, T, I ratings 0 to 5; quality band Q0 to Q5.
sufficiency	Sufficiency S0 to S5 for the assertion concerned.
restrictions	DistributionControl reference.

Assessment profile

Field	Content
programId	AssuranceProgram.
decisionQuestion	The question the assessment answers.
capabilityId, scope	CapabilityInstance, ServiceOutcomes, boundary, ArchitectureSnapshot, states.
profile	Capability profile, parameter set and weighting (if any).
team	Assessors and specialists, with independence declarations.
evidenceManifest	EvidencePackage references.
results, gates	ControlAssessments and GateResults (COMPOSED_OF).
review	ReviewRecords.

Finding profile

Field	Content
criterionRef	Control and criterion (or gate criterion) not met.
condition, cause, consequence	What was found, why, and its effect on outcomes.
evidenceRefs	Evidence of the condition.
classification	Finding class (finding-classes registry).
gateEffect, scoreEffect	Effect on GateResults and ScoreSnapshot.
owner, status	Accountable owner and lifecycle.

UNKNOWN profile (UnknownRecord)

Field	Content
subjectRef	Record or attribute that is unknown.
unknownType	U1 to U10 (AIDCF-008).
severity	US1 to US5 (canonical specification §5.7).
reason	Why it cannot be determined.
decisionEffect	Ceiling or gate effect implied by the severity.
evidenceNeeded	Evidence that would resolve it.
owner, trigger, status	Owner, resolution trigger and lifecycle.

GateResult profile

Field	Content
gateId	CG-01 to CG-15.
applicability	In scope, or APPROVED_NA with boundary rationale and approver.
criticalInputs	ControlAssessments of CRITICAL controls and their results.
supportingInputs	ControlAssessments of SUPPORTING controls; gate-level evaluation for each NONCONFORMING.
criteria	Gate criteria evaluated and outcome.
evidenceRefs	Gate evidence and achieved sufficiency against the floor.
stateCoverage	OperatingStates covered.
findings, unknowns	Linked Findings and UnknownRecords.
result	Gate result code (§11A.4).
decisionEffect	Ceiling applied to the decision (AIDCF-010).

Score profile (ScoreSnapshot)

Field	Content
subjectRef	Assessment and scope.
profileVersion	Scoring profile and weighting version.
inputs	Pinned ControlAssessment versions.
values	Control score values (100, 50, 0; UNKNOWN excluded) and family/domain means.
coverage	Scored / (in-scope − NOT_APPLICABLE), per family and domain.
band	B1 to B5, or "Band withheld (coverage/UNKNOWN)".
unknownOverlay	UNKNOWN counts by severity.

Decision profile (AssuranceDecision)

Field	Content
question, subjectRef, scope	Decision question, CapabilityInstance and scope.
recommendationRef	DecisionRecommendation (RECOMMENDS).
gateResults, unknowns, scores	Inputs, referenced directly.
conditions	Conditions and limitations, each with owner and due date.
decisionState	Decision state (§11A.5).
lifecycle	Decision lifecycle (§11A.6).
authority	ReviewRecord of the approving authority (APPROVED_BY).
validity	effectiveFrom and effectiveTo; reassessment triggers.

Report profile (AssuranceReport)

Field	Content
reportClass	Report type (report-types registry, AIDCF-012).
programRef, assessmentRef, decisionRef	Lineage.
issuer, users, intendedUse	Issuer, intended users and permitted use.
statements	AssuranceStatements (REPORTED_IN).
validity, approvals	Validity period and ReviewRecords.

Event profile

Field	Content
eventType	Event type code (§21).
subjectRef, source	Affected entity and reporting source with its health.
observedAt, effectiveAt	When observed and when it took effect.
payload	Type-specific content, schema-versioned.
severity, materiality	Assessed significance (candidate until reviewed).
triggerRule	Matched ReassessmentTrigger.
affectedAssertions	Records identified by impact analysis (§22).
reviewState	Human disposition state.

9 Relationship registry

TDS-REQ-039 The 41 relationship types below are the canonical relationship vocabulary of AIDCF. Every edge in an AIDCF graph shall use one of them (or a registered extension, §33), shall respect the stated domain and range, and shall be validated against the stated cardinality.

Conventions:

- **Domain** is the class of the edge's source; **range** is the class of its target. "Architecture element" means InfrastructureSystem, AssetInstance, Interface or Resource.
- **Cardinality** is written *out* / *in*: *out* is the number of edges of this type that each source instance may have; *in* is the number each target instance may have. Unless otherwise stated, cardinality is evaluated within one ArchitectureSnapshot or one Assessment.

- **Inverse** names the inverse property defined in the ontology for query and navigation. Inverses are derived, never asserted separately, so that an edge is stored once.
- All edges carry the relationship envelope (§10).

#	Type	Definition	Domain	Range	Cardinality (out / in)	Inverse
1	HAS_OUTCOME	The capability instance is required to achieve the target service outcome.	CapabilityInstance	ServiceOutcome	1..* / 1..1	outcomeOf
2	REQUIRES_FUNCTION	Achieving the outcome requires the target function to be delivered.	ServiceOutcome	RequiredFunction	1..* / 1..*	functionRequiredBy
3	IMPLEMENTED_BY	The function, or the safeguard, is realised by the target architecture element.	RequiredFunction, Safeguard	InfrastructureSystem, AssetInstance, Interface	1..* / 0..*	implements
4	COMPOSED_OF	Whole-part structure: the source contains the target as a constituent. Acyclic.	ArchitectureSnapshot, InfrastructureSystem, AssetInstance, ControlDomain, ControlFamily, AssuranceProgram, Assessment, EvidencePackage	InfrastructureSystem, AssetInstance, Component, Interface, Resource, ControlFamily, Control, Assessment, ControlAssessment, GateResult, Sample, TestRecord, EvidenceObject	0..* / 0..* (ControlFamily in 1..1 from ControlDomain; Control in 1..1 from ControlFamily)	partOf
5	CONNECTS_TO	The source is an endpoint of the target interface. Physical or logical connection only; implies no dependency, causality or reachability.	InfrastructureSystem, AssetInstance	Interface	0..* / 2..*	hasEndpoint
6	CONSUMES_RESOURCE	The source draws on the target constrained resource (power, cooling, water, fuel, space, accelerator capacity).	RequiredFunction, InfrastructureSystem, AssetInstance	Resource	0..* / 0..*	consumedBy
7	DEPENDS_ON	The source relies on the target to perform its function; each edge references exactly one Dependency record carrying type, strength and states. AIDCF-003 specialised predicates (POWERED_BY, COOLED_BY, SUPPLIED_BY, AUTHENTICATED_BY, TIMED_BY, CONFIGURED_BY, MAINTAINED_BY) are DEPENDS_ON with the corresponding dependencyType.	RequiredFunction, Safeguard, architecture element	Architecture element, Safeguard, PartyEntity	0..* / 0..*	dependencyOf
8	SHARES_FAULT_DOMAIN_WITH	Source and target are members of the same FaultDomain and can fail together from its common cause. Symmetric; each edge references the FaultDomain record.	Architecture element, Safeguard	Architecture element, Safeguard	0..* / 0..*	(symmetric)
9	LOCATED_IN	The source is physically or logically placed in the target location.	InfrastructureSystem, AssetInstance, FaultDomain	Location	0..1 / 0..*	locationOf
10	OPERATED_BY	The target party holds operational responsibility for the source.	CapabilityInstance, InfrastructureSystem, AssetInstance, Safeguard	PartyEntity	1..* / 0..*	operates

#	Type	Definition	Domain	Range	Cardinality (out / in)	Inverse
11	CONTROLLED_BY	The target control system or actor can command the source's state or setpoints (for example a CDU controlled by a BMS, a node controlled by a cluster manager via BMC).	InfrastructureSystem, AssetInstance	InfrastructureSystem, AssetInstance, PartyEntity	0..* / 0..*	controls
12	MONITORED_BY	The target observes the source and is relied on to detect its condition.	Architecture element, Safeguard, Dependency	InfrastructureSystem, AssetInstance	0..* / 0..*	monitors
13	PROTECTED_BY	The target safeguard acts to prevent, detect, contain, correct or recover from failure of, or harm to, the source.	ServiceOutcome, RequiredFunction, architecture element	Safeguard	0..* / 1..*	protects
14	EXPOSED_TO	The source is credibly susceptible to the target threat, hazard or failure mode.	CapabilityInstance, architecture element, Safeguard	Threat, Hazard, FailureMode	0..* / 0..*	exposes
15	INITIATES	The threat, hazard or failure mode is the initiating source of the target scenario.	Threat, Hazard, FailureMode	FailureScenario	0..* / 1..*	initiatedBy
16	PROPAGATES_TO	Within a named scenario, failure of the source causes failure or degradation of the target; carries scenarioRef and step ordinal.	Architecture element, RequiredFunction	Architecture element, RequiredFunction, ServiceOutcome	0..* / 0..*	propagatedFrom
17	RECOVERS_THROUGH	The source is restored to tolerance by means of the target recovery path, alternate capacity or recovery safeguard.	ServiceOutcome, RequiredFunction, architecture element	Architecture element, Safeguard	0..* / 0..*	recoveryPathFor
18	GOVERNED_BY	The source is subject to the requirements, method or disclosure rule of the target.	Safeguard, ControlAssessment, Assessment, any record (for DistributionControl)	Control, AssessmentProcedure, AssuranceProgram, DistributionControl	0..* / 0..* (Assessment out 1..1 to AssuranceProgram)	governs
19	HAS_EVIDENCE_REQUIREMENT	The control or gate requires the target evidence requirement for a stated claim type.	Control, CriticalGate	EvidenceRequirement	1..* / 1..1	requirementOf
20	EVIDENCED_BY	The source is offered the target evidence. Records that evidence was presented, not that it is sufficient.	EvidenceRequirement, CapacityAssertion, Dependency, FaultDomain, ControlAssessment, GateResult, Finding	EvidenceObject, EvidencePackage	0..* / 0..*	evidences
21	SUPPORTS	After review, the source evidence corroborates the target claim.	EvidenceObject	CapacityAssertion, Dependency, FaultDomain, ControlAssessment, GateResult, FailureScenario	0..* / 0..*	supportedBy
22	CHALLENGES	The source contradicts or undermines the target claim (defeater). Alias: DISPUTES.	EvidenceObject, Finding, UnknownRecord, ReviewRecord, BaselineIssue	Any assertion-bearing record	0..* / 0..*	challengedBy
23	QUALIFIES	The source limits the scope, states or conditions under which the target claim holds, without contradicting it.	EvidenceObject, ReviewRecord, UnknownRecord	Any assertion-bearing record	0..* / 0..*	qualifiedBy

#	Type	Definition	Domain	Range	Cardinality (out / in)	Inverse
24	ASSESSED_BY	The control is assessed, for one claim type in one scope, by the target control assessment.	Control	ControlAssessment	0..* / 1..1	assesses
25	RESULTS_IN	Execution of the procedure, sample or test produced (contributes to) the target control assessment.	AssessmentProcedure, Sample, TestRecord	ControlAssessment	0..* / 1..*	resultOf
26	HAS_FINDING	The source result or gate evaluation records the target finding.	ControlAssessment, GateResult, Assessment	Finding	0..* / 1..*	findingOf
27	HAS_UNKNOWN	The source has a material undetermined fact recorded by the target UNKNOWN record.	Any record	UnknownRecord	0..* / 1..*	unknownOf
28	EVALUATES_GATE	Control to CriticalGate: registry gate mapping, mandatory attribute gateRole (CRITICAL or SUPPORTING). ControlAssessment to GateResult: the result is an input to that gate evaluation. GateResult to CriticalGate: the result evaluates that gate.	Control, ControlAssessment, GateResult	CriticalGate, GateResult	Control 0..* ; GateResult out 1..1 / CriticalGate in 0..*	evaluatedBy
29	CONSTRAINS	The source imposes a ceiling, condition, limitation or band withholding on the target. Attribute constraintType: CEILING, CONDITION, LIMITATION, BAND_WITHHELD, GATE_EFFECT. Aliases: CONSTRAINS_GATE, LIMITS_DECISION.	GateResult, UnknownRecord, Finding	GateResult, ScoreSnapshot, DecisionRecommendation, AssuranceDecision, AssuranceStatement	0..* / 0..*	constrainedBy
30	SUMMARISED_BY	The source results are aggregated into the target score snapshot or recommendation.	ControlAssessment, GateResult, ScoreSnapshot, UnknownRecord	ScoreSnapshot, DecisionRecommendation	0..* / 1..*	summarises
31	RECOMMENDS	The recommendation proposes the decision state and conditions taken up (or departed from, with reasons) by the target decision.	DecisionRecommendation	AssuranceDecision	0..1 / 1..1	recommendedBy
32	REVIEWED_BY	The source record has been reviewed in the target review record.	Any record	ReviewRecord	0..* / 1..*	reviews
33	APPROVED_BY	The source record has been approved by the authority recorded in the target review record (reviewType APPROVAL).	AssuranceDecision, AssuranceStatement, AssuranceReport, ReleaseManifest, MigrationRecord, ArchitectureSnapshot, BaselineIssue resolution, Control (registry release)	ReviewRecord	0..* / 1..* (AssuranceDecision out 1..*)	approves
34	REPORTED_IN	The source is disclosed in the target report.	AssuranceDecision, AssuranceStatement, Finding, UnknownRecord, GateResult, ScoreSnapshot	AssuranceReport	0..* / 0..*	reports

#	Type	Definition	Domain	Range	Cardinality (out / in)	Inverse
35	REQUIRES_ACTION	The source requires the target action to be completed (remediation, evidence collection, decision condition).	Finding, UnknownRecord, AssuranceDecision, BaselineIssue	Action	0..* / 1..*	actionFor
36	CLOSED_BY	The source is closed by the target closure evidence, reassessment result or review.	Finding, UnknownRecord, Action, BaselineIssue	ReviewRecord, EvidenceObject, ControlAssessment, MigrationRecord	0..* / 0..*	closes
37	INVALIDATED_BY	The source can no longer be relied on because of the target event (evidence expiry, configuration drift, incident, test failure).	EvidenceObject, CapacityAssertion, ControlAssessment, GateResult, ScoreSnapshot, AssuranceDecision, AssuranceStatement	Event	0..* / 0..*	invalidates
38	TRIGGERS_REASSESSMENT	The source event or trigger rule requires reassessment of the target.	Event, ReassessmentTrigger	Control, ControlAssessment, GateResult, Assessment, AssuranceDecision, ServiceOutcome	0..* / 0..*	reassessmentTriggeredBy
39	SUPERSEDES	The source replaces the target record of the same class; the target is retained with status SUPERSEDED. Acyclic.	Any record	Record of the same class	0..* / 0..1	supersededBy
40	MIGRATED_FROM	The source was produced from the target (prior version, legacy vocabulary or legacy record) by the migration recorded in migrationRef.	Any record	Any record	0..* / 0..*	migratedTo
41	INTEGRATION_HANDOFF	The source AIDCF record is exchanged with the AI Trust Graph through the target reviewed handoff record (AIDCF-005).	Any canonical record	IntegrationHandoff	0..* / 1..*	handoffOf

9.1 Legacy relationship aliases

TDS-REQ-040 Legacy relationship names shall be accepted only on input and shall be migrated by a MigrationRecord to the canonical type, reversing direction where stated. They shall not be emitted.

Legacy name	Canonical type	Direction change
SUPPORTS_OUTCOME	IMPLEMENTED_BY	Reverse: "System SUPPORTS_OUTCOME Function" becomes "Function IMPLEMENTED_BY System". Outcome links use HAS_OUTCOME and REQUIRES_FUNCTION.
IMPLEMENTS	IMPLEMENTED_BY	Reverse.
CONSTRAINS_GATE	CONSTRAINS	None; constraintType GATE_EFFECT.
LIMITS_DECISION	CONSTRAINS	None; constraintType CEILING or LIMITATION.
PROTECTS	PROTECTED_BY	Reverse.
DISPUTES	CHALLENGES	None.
EVALUATES	ASSESSED_BY	Reverse ("ControlAssessment EVALUATES Control" becomes "Control ASSESSED_BY ControlAssessment").
TESTED_BY	EVIDENCED_BY	None; target is a TestRecord-derived EvidenceObject.
RECOVERED_BY	RECOVERS_THROUGH	None.

10 Relationship envelope

TDS-REQ-041 Every edge shall be a first-class record carrying the fields below. Edges without a relationshipId, type, sourceId and targetId are invalid.

Field	Rule
relationshipId	Stable identifier; edges are versioned like records.
type	One of the 41 types or a registered extension.
sourceId, targetId	Identifiers whose classes satisfy the domain and range in §9.
semanticOwner	Owner artifact of the relationship type.
scope	ArchitectureSnapshot or Assessment within which the edge is asserted.
conditions	OperatingStates and other conditions under which the edge holds (mandatory for DEPENDS_ON, PROTECTED_BY, PROPAGATES_TO, RECOVERS_THROUGH).
qualifiers	Type-specific attributes: dependencyRef, faultDomainRef, scenarioRef and step, gateRole, constraintType, migrationRef.
effectiveFrom, effectiveTo	Validity interval.
sourceRecord	Originating record in a source system, where imported.
evidenceRefs	Evidence for the edge itself (for example independence evidence for the absence of a shared fault domain).
confidence	CONF-0 to CONF-4, with CONFLICTING and STALE flags.
reviewState	CANDIDATE until reviewed (§11A.2).
classification	Classification code.
version, supersedes	Version and superseded edge identifiers.

11 End-to-end traceability

TDS-REQ-042 A conformant dataset at T2 or above shall evaluate every rule below for every in-scope record and shall report each failure as a ValidationResult. A traceability gap that cannot be closed by data shall be represented as an UnknownRecord on the source record; it shall not be ignored.

Rule	From → to	Test	Chain steps
TR-01	Capability → Outcome	Every CapabilityInstance has ≥1 HAS_OUTCOME; every ServiceOutcome has exactly one incoming HAS_OUTCOME.	2→3
TR-02	Outcome → Function	Every ServiceOutcome has ≥1 REQUIRES_FUNCTION.	3→4
TR-03	Function → Architecture	Every RequiredFunction has ≥1 IMPLEMENTED_BY to an element in the current ArchitectureSnapshot.	4→5
TR-04	Architecture → Dependency	Every element implementing a RequiredFunction has its DEPENDS_ON edges recorded, or an UnknownRecord stating that dependency discovery is incomplete.	5→6
TR-05	Dependency → Failure	Every Dependency of strength ABSOLUTE or MATERIAL, and every FaultDomain containing redundant elements, is covered by ≥1 FailureScenario.	6→7
TR-06	Failure → Safeguard	Every FailureScenario whose outcome effect exceeds tolerance identifies the Safeguards expected to act (PROTECTED_BY on the affected elements).	7→8
TR-07	Safeguard → Control	Every Safeguard relied on in a scenario is GOVERNED_BY ≥1 Control.	8→9
TR-08	Control → Evidence Requirement	Every in-scope Control has HAS_EVIDENCE_REQUIREMENT for its claim type.	9→10
TR-09	Requirement → Evidence or UNKNOWN	Every EvidenceRequirement in scope has EVIDENCED_BY ≥1 EvidenceObject meeting its floors, or HAS_UNKNOWN.	10→11
TR-10	Procedure → Result	Every ControlAssessment has ≥1 RESULTS_IN from a procedure, sample or test, names its method codes and assessor, and is REVIEWED_BY.	11→12
TR-11	Result → Finding or UNKNOWN	Every PARTIAL or NONCONFORMING result has HAS_FINDING; every UNKNOWN result has HAS_UNKNOWN.	12→13
TR-12	Gate → Evidence, Findings, UNKNOWN	Every GateResult has EVALUATES_GATE to its CriticalGate, links the ControlAssessments of all in-scope CRITICAL controls, and links its evidence, Findings and UnknownRecords.	13→14
TR-13	Score → immutable inputs and profile	Every ScoreSnapshot references pinned input versions and profile version, and recomputation from them yields the same values.	12→15
TR-14	Recommendation → constraints	Every DecisionRecommendation links all GateResults and every US3–US5 UnknownRecord in scope and applies their CONSTRAINS edges.	14→15
TR-15	Decision → authority, scope, validity	Every AssuranceDecision has RECOMMENDS incoming, APPROVED_BY a ReviewRecord of the declared authority, a scope and effectiveTo.	15→16
TR-16	Report → approved decision	Every AssuranceStatement and AssuranceReport traces to an AssuranceDecision whose lifecycle is APPROVED or ACTIVE, and does not exceed it.	16→Loop
TR-17	Trigger → impacted objects	Every Event matched to a ReassessmentTrigger yields TRIGGERS_REASSESSMENT or INVALIDATED_BY edges to all dependent records found by §22.	Loop→2

Rule	From → to	Test	Chain steps
TR-18	Supersession → prior and successor	Every SUPERSEDED record has exactly one supersededBy and the chain is acyclic.	All

11A Controlled vocabularies

TDS-REQ-043 The codes in this section and in the canonical specification are the only codes that conformant implementations shall emit. Codes are published as CSV files under `registries/vocabularies/` (one file per vocabulary) and listed in the ReleaseManifest.

TDS-REQ-044 Legacy codes shall be accepted only on input, shall be converted by a MigrationRecord using the alias tables below, and the original code shall be preserved in provenance.

Vocabularies defined elsewhere and used unchanged: claim types and assurance levels AL0–AL6 (AIDCF-011 and the canonical specification §5.1), confidence CONF-0 to CONF-4 (AIDCF-008), evidence classes E1–E7, quality dimensions, bands Q0–Q5, sufficiency S0–S5, integrity I0–I5, profile tiers CP-1 to CP-5 and evidence-system conformance EC0–EC5 (AIDCF-008), UNKNOWN types U1–U10 (AIDCF-008) and severities US1–US5, assessment methods, control types, lifecycle stages and score bands B1–B5 (AIDCF-009, AIDCF-010).

11A.1 Record lifecycle

Applies to the `status` field of every record (not to decision state).

Code	Meaning
DRAFT	Being prepared; not for use in assessment.
CANDIDATE	Proposed (often by automation); awaiting review.
REVIEWED	Reviewed; not yet approved.
APPROVED	Approved by the accountable authority.
ACTIVE	Approved and in use.
SUSPENDED	Temporarily not to be relied on, pending reassessment.
WITHDRAWN	Removed from use before expiry; retained for history.
SUPERSEDED	Replaced by a newer record (SUPERSEDES).
RETIRED	End of life without replacement; retained for history.

11A.2 Assertion review state

Code	Meaning
CANDIDATE	Asserted but not reviewed.
SUPPORTED	Reviewed; supported by evidence meeting the applicable floor.
CHALLENGED	Subject to an unresolved CHALLENGES edge.
QUALIFIED	Holds only under the conditions stated by QUALIFIES edges.
CONFLICTING	Supported and challenged by evidence of comparable quality.
UNKNOWN	Cannot be determined; HAS_UNKNOWN is mandatory.
SUPERSEDED	Replaced by a newer assertion.

11A.3 Control result (ControlAssessment.result)

Code	Meaning	Score value
CONFORMING	All assessment criteria met for the claim type; evidence at or above the profile floor.	100
PARTIAL	Some criteria met; specific deficiencies recorded as Findings.	50
NONCONFORMING	Criteria materially not met, or a safeguard demonstrated ineffective.	0
UNKNOWN	Result cannot be determined (evidence below floor, conflict, inaccessible).	None; excluded from mean; counted in coverage and the UNKNOWN register
NOT_APPLICABLE	Approved non-applicability with rationale and approver.	Excluded
NOT_ASSESSED	In scope but not yet assessed.	Excluded; counted as coverage gap

Legacy alias table:

Legacy code	Canonical code	Condition
NOT_IMPLEMENTED	NONCONFORMING	None.
INEFFECTIVE	NONCONFORMING	None.
IMPLEMENTED	CONFORMING	Valid only for DESIGN or IMPLEMENTATION claims; for any other claim type the migrated result is UNKNOWN.
OPERATING_EFFECTIVELY, STRONGLY_EFFECTIVE	CONFORMING	For OPERATION or EFFECTIVENESS claims.
N/A	NOT_APPLICABLE	Rationale and approver must exist, otherwise the migrated result is UNKNOWN.
Not Assessed	NOT_ASSESSED	None.
SUSPENDED	UNKNOWN	Record lifecycle of the ControlAssessment set to SUSPENDED.

Implementation maturity IM0 to IM5 (AIDCF-017) is a diagnostic attribute and is never a result code.

11A.4 Gate result (GateResult.result)

Code	Meaning
PASS	All CRITICAL controls CONFORMING at required claim type, gate criteria met, evidence at floor, SUPPORTING nonconformities evaluated.
FAIL	Any in-scope CRITICAL control NONCONFORMING or PARTIAL, or a gate criterion demonstrated not met.
UNKNOWN	Not FAIL, and any CRITICAL control UNKNOWN or NOT_ASSESSED, gate evidence below floor, or a US4 UNKNOWN attached.
APPROVED_NA	Gate out of scope with approved boundary rationale.
NOT_ASSESSED	Gate in scope but not yet evaluated.

There is no conditional gate result; conditions exist only at decision level. Legacy spelling `APPROVED_N/A` is an input alias of APPROVED_NA.

11A.5 Decision state (AssuranceDecision.decisionState)

Code	Meaning (rules in AIDCF-010)
REASONABLY_ASSURED	All applicable gates PASS; no US3–US5 UNKNOWN open; scores and coverage support the claim.
CONDITIONALLY_ASSURED	Gates PASS; specified conditions with owners and dates must be met for the claim to hold.
LIMITED	Assurance holds only for a reduced scope, set of outcomes or states, stated explicitly.
NOT_ASSURED	Evidence shows the claim does not hold (for example a gate FAIL that defeats it).
INCONCLUSIVE	The claim cannot be decided (for example US5 UNKNOWN or gate UNKNOWN beyond ceiling).

Legacy alias table:

Legacy code	Canonical code
ADVERSE, ADVERSE_NOT_ASSURED, "Adverse / Not assured"	NOT_ASSURED
UNKNOWN, UNKNOWN_INCONCLUSIVE, UNKNOWN_OR_INCONCLUSIVE, Withheld	INCONCLUSIVE
CONDITIONAL	CONDITIONALLY_ASSURED
SUSPENDED, WITHDRAWN, SUPERSEDED (formerly listed as states)	Decision lifecycle codes (§11A.6); decision state unchanged
NOT ASSESSED	Not a decision state; no AssuranceDecision is issued

11A.6 Decision lifecycle (AssuranceDecision.lifecycle)

Code	Meaning
DRAFT	Being prepared.
RECOMMENDED	DecisionRecommendation complete; awaiting authority.
APPROVED	Approved by the authority; not yet effective.
ACTIVE	In effect within its validity period.
SUSPENDED	Not to be relied on pending reassessment after a triggering event.
WITHDRAWN	Revoked by the authority.
SUPERSEDED	Replaced by a newer decision.
EXPIRED	Validity period ended without renewal.

11A.7 Operating state

Owner AIDCF-003 §45.

Code	Meaning
NORMAL	Approved architecture operates within declared envelope; no material planned impairment.
DEGRADED	One or more functions, safeguards, paths or reserves are impaired while some capability remains.
MAINTENANCE	A planned intervention changes availability, configuration, staffing or safeguards.
EMERGENCY	Exceptional condition requiring urgent protective action.
RECOVERY	Controlled restoration and validation after disruption.
TRANSITION	Movement between architectures, states, locations or control regimes.
SHUTDOWN	Intentional cessation with controlled residual dependencies.
FAILED	Required function is not delivered.
UNKNOWN	Current operating state cannot be determined.

11A.8 Classification

Code	Meaning
PUBLIC	Approved for public release.
INTERNAL	For the owning organisation and its assessors.
CLIENT_RESTRICTED	For named client parties under agreement.
CONFIDENTIAL	Need-to-know within named roles; disclosure would cause commercial or security harm.
HIGHLY_RESTRICTED	Exploitable detail (for example OT access paths, protection settings); named individuals only.

12 Serialisation architecture

TDS-REQ-045 AIDCF-016 is the normative serialisation companion to this standard. It defines the schema and ontology package (JSON Schema, JSON-LD context, OWL ontology, SHACL shapes, CSV layouts) that realises §7 to §11A. Conformant implementations shall serialise AIDCF data as defined in AIDCF-016 for the release pinned in the ReleaseManifest.

Format	Role	Authority
JSON	Application exchange of records and edges	Authoritative record form
JSON Schema 2020-12	Structural validation	Normative (AIDCF-016)
JSON-LD 1.1	Mapping of JSON keys to `aidcf:` IRIs	Normative (AIDCF-016)
RDF (Turtle)	Graph exchange	Equivalent projection of JSON records
OWL 2	Class, property, domain, range and inverse declarations	Normative (AIDCF-016)
SHACL	Closed-world cardinality and business-rule validation	Normative (AIDCF-016)
CSV	Registries and controlled vocabularies	Normative for codes
NDJSON	Event streams	Exchange only
Parquet or similar columnar	Analytics	Non-authoritative

13 JSON design rules

TDS-REQ-046 JSON records shall comply with every rule below; AIDCF-016 schemas enforce them.

Rule	Requirement
Encoding	UTF-8 without byte-order mark.
Naming	Property names in lowerCamelCase; class names in UpperCamelCase; codes in UPPER_SNAKE_CASE.
Dates and times	ISO 8601 with explicit UTC offset; durations as ISO 8601 durations.
Explicit null reason	A null value is permitted only with a sibling `nullReason` (NOT_APPLICABLE with rationale reference, or UNKNOWN with UnknownRecord reference).
Enumerated codes	Every coded field is an enum bound to a vocabulary file version.
Stable schema identifiers	Each schema has a versioned `$id`; records declare the schema version they conform to.
additionalProperties false	Every object schema sets `additionalProperties: false`. Extension data is permitted only under `extensions` (§33).
Namespaced extensions	Keys under `extensions` are registered namespace prefixes.
Discriminator	Every record carries `type`; polymorphic references carry the target type.
Stable references	References are identifiers, never array positions or display names.

Rule	Requirement
Integrity metadata	Records exported in packages carry a digest; packages carry a manifest of digests.
Classification	Every record carries `classification`.
Units	Quantities carry value and unit; units are taken from a declared unit vocabulary (SI and derived units).

14 Semantic graph rules

Requirement	Rule
TDS-REQ-047	JSON-LD contexts shall use protected term definitions so that an imported context cannot redefine AIDCF terms.
TDS-REQ-048	The `aidcf:` prefix shall resolve to `https://w3id.org/aidcf/ontology#`; instance IRIs follow the AIDCF-016 pattern.
TDS-REQ-049	AIDCF and AI Trust Graph terms shall be held in separate contexts and named graphs; mixing is permitted only through IntegrationHandoff records.
TDS-REQ-050	OWL classes and properties shall mirror §7 and §9 exactly, including declared inverses and symmetric properties.
TDS-REQ-051	OWL or rule-based inference shall not produce APPROVED, SUPPORTED, CONFORMING, PASS or any decision state; inferred triples are held in a separate named graph with reviewState CANDIDATE.
TDS-REQ-052	Cardinality, mandatory-link and vocabulary constraints shall be validated with SHACL under closed-world assumptions, not inferred under OWL open-world semantics.
TDS-REQ-053	Ontology imports shall be pinned by version IRI.
TDS-REQ-054	Deprecated terms shall remain declared with `owl:deprecated true` and a pointer to their replacement.

15 JSON Schema package

The schema package is defined, versioned and published by AIDCF-016. The logical coverage it must provide is listed below; AIDCF-016 may split or combine files provided coverage is complete.

TDS-REQ-055 The AIDCF-016 schema package shall provide schemas covering, at minimum: release manifest; common envelope and relationship envelope; capability, outcome and function; architecture snapshot, system, asset, interface, resource and capacity assertion; dependency and fault domain; threat, hazard, failure mode and failure scenario; safeguard; control domain, family and control; evidence requirement, object and package; assurance programme, assessment, procedure, sample and test record; control assessment, finding and UNKNOWN record; critical gate and gate result; score snapshot; decision recommendation, review record and assurance decision; assurance statement and report; action, reassessment trigger and event; integration handoff; validation result, migration record, baseline issue and distribution control.

AIDCF-016 v1.0 publishes the MDM 1.3 schema set in `schemas/json-schema/`: `definitions.schema.json`, `common-envelope.schema.json` and one closed JSON Schema 2020-12 file per class (58 classes: the 48 canonical classes of §7, the 3 reference classes, 6 supporting classes pending registration and the auxiliary RelationshipRecord). The set is generated from `schemas/tools/aidcf_model.py` by `schemas/tools/build_schemas.py`, whose `--check` mode fails if a published file differs from the generated one. The earlier v1.2 subset (control assessment, evidence, gate, score, decision) is superseded. Its only copies are those released with the AIDCF-015 v1.2 package under `implementation/aidcf-015-v1.2/schemas/`. They are kept unchanged for that package, and data validated against them is migrated to MDM 1.3 under §27.

16 Semantic publication package

TDS-REQ-056 AIDCF-016 shall publish, for each release, a JSON-LD context, an OWL ontology with version IRI, a Turtle serialisation, SHACL shapes, the registry and vocabulary CSV files, example instance data, and the conformance-test fixtures, each listed with its digest in the ReleaseManifest. File names and versions are defined by AIDCF-016.

The current release, MDM 1.3 (AIDCF-016 v1.0), publishes the JSON-LD context `schemas/jsonld/aidcf-context-1.3.jsonld`, the OWL ontology with SKOS concept schemes `schemas/owl/aidcf-core-1.3.ttl` (Turtle; version IRI `https://w3id.org/aidcf/ontology/1.3`) and the SHACL shapes `schemas/shacl/aidcf-shapes-1.3.ttl`. The registries and vocabularies are the CSV files under `registries/`, and the example data and conformance fixtures are in the AIDCF-015 v1.3 package (`implementation/aidcf-015-v1.3/`). w3id registration is pending, so the IRIs do not yet resolve. The v1.2 files (`aidcf-context-v1.2.jsonld`, `aidcf-core-v1.2.ttl`, `aidcf-shapes-v1.2.ttl`) are superseded and are no longer under `schemas/`. They survive only in `implementation/aidcf-015-v1.2/semantic/`, unchanged as released with that package.

17 Registry catalogue

TDS-REQ-057 Every registry CSV shall carry the common columns `code`, `label`, `definition`, `semanticOwner`, `version`, `status` and `replacedBy`, plus the registry-specific columns below.

Registry file	Registry-specific columns	Owner
artifacts.csv	publicId, shortCode, title, path, artifactVersion, artifactStatus	AIDCF-014
object-types.csv	chainStep, ownerArtifact, schemaRef	AIDCF-014
relationship-types.csv	domain, range, cardinalityOut, cardinalityIn, inverse, symmetric, qualifiers	AIDCF-014
vocabularies/*.csv	vocabulary, sortOrder, scoreValue (where applicable), legacyAliases	Per §11A
domains.csv	familyCount, controlCount	AIDCF-007A
control-families.csv	domainCode, familyNumber, controlRange	AIDCF-007A
controls.csv	familyId, claimType, requiredAssuranceLevel, controlType, gateMappings (gateId:role list), amendmentRef	AIDCF-007A
critical-gates.csv	name, scope, evidenceFloor, criticalControlCount	AIDCF-010
evidence-classes.csv	classCode, examples	AIDCF-008
evidence-quality.csv	dimension, level, criterion	AIDCF-008
assessment-methods.csv	methodCode, typicalEvidenceClasses, safetyConstraint	AIDCF-009
finding-classes.csv	severity, gateEffect	AIDCF-009
unknown-types.csv	typeCode (U1–U10), defaultSeverityTest	AIDCF-008
unknown-severity.csv	severityCode (US1–US5), test, decisionEffect	AIDCF-010
decision-states.csv	rank, ceilingRule	AIDCF-010
report-types.csv	audience, mandatorySections	AIDCF-012
architecture-patterns.csv	patternCode, mandatoryViews, typicalGates	AIDCF-013
external-mappings.csv	externalScheme, externalCode, aidcfCode, mappingType, crosswalkRef	AIDCF-019
deprecated-terms.csv	deprecatedTerm, canonicalTerm, directionChange, effectiveRelease	AIDCF-014
migration-rules.csv	sourceVersion, targetVersion, rule, condition, fallback	AIDCF-014

Implementation status. `registries/catalogue/registry-index.csv` lists the twenty registries above with their published path, owner, row count, sources, whether the common columns are present, and the columns still missing. The index is computed from the published file headers. `registries/tools/build_catalogue.py` generates fifteen of the registries in `registries/catalogue/` with the full common and registry-specific columns: artifacts, object types, relationship types, evidence classes, evidence quality, assessment methods, finding classes, UNKNOWN types, UNKNOWN severity, decision states, report types, architecture patterns, external mappings, deprecated terms and migration rules. The tool's `--check` mode fails if regenerating would change any file. Where no source artifact defines a value, the cell is left empty rather than inferred. Examples are AIDCF-013 pattern codes, finding-class severity, typical evidence classes per assessment method, report-type definitions and effective releases of deprecated terms.

The UNKNOWN-severity entry above was listed in earlier drafts as `unknown-severities.csv`. It is now named `unknown-severity.csv`, the vocabulary file name. The catalogue file of that name is a projection of `registries/vocabularies/unknown-severity.csv`, and if the two differ the vocabulary file governs. The same holds for the catalogue projections of evidence classes, UNKNOWN types, decision states and assessment methods.

TDS-REQ-057 is not yet met in full. The domains, control-families, controls and critical-gates registries and the 25 vocabulary files are maintained by other tools, and they do not yet carry all of the common columns. For example, they use `domainId`, `familyId`, `controlId`, `gateId` or `name` in place of `code` and `label`, and they have no `semanticOwner` or `replacedBy`. Their registry status `PROPOSED` is also outside the §11A.1 record lifecycle. The missing columns of each are listed in `registry-index.csv`. A registry is conformant only when its index row shows no missing columns.

TDS-REQ-058 The controls registry shall contain exactly 360 rows, 24 per domain and four per family; the families registry exactly 90 rows; the domains registry exactly 15 rows. Any deviation is a BLOCKED BaselineIssue.

18 Business rules

TDS-REQ-059 Implementations claiming T5 or above shall enforce the business rules below and shall reject or flag (as stated) records that violate them.

Rule	Statement	On violation
BR-001	A ControlAssessment shall identify its control, claim type, scope subject and the applicability basis for that subject.	Reject
BR-002	NOT_APPLICABLE requires a rationale and an approving ReviewRecord. Missing evidence yields UNKNOWN, never NOT_APPLICABLE.	Reject
BR-003	A CONFORMING result applies only to the claim type assessed; it shall not be presented as conformance for a higher claim type.	Reject
BR-004	A GateResult of PASS requires evidence at or above the gate evidence floor and CONFORMING results for every in-scope CRITICAL control.	Reject
BR-005	REASONABLY_ASSURED is not permitted while any applicable GateResult is FAIL or UNKNOWN, or any US4/US5 UnknownRecord is open in scope.	Reject
BR-006	A ScoreSnapshot shall state coverage and UNKNOWN overlay; a band shall be withheld when coverage < 80% or a US4/US5 UNKNOWN is open in scope. A score never changes a gate.	Reject

Rule	Statement	On violation
BR-007	An AssuranceStatement or AssuranceReport shall not state a decision state, scope, outcome or validity stronger than the AssuranceDecision it reports.	Reject
BR-008	Closure of a Finding or UnknownRecord requires CLOSED_BY closure evidence or a reassessment result, reviewed by someone other than the action owner where independence is required.	Reject
BR-009	Supersession preserves history: superseded records remain retrievable and unchanged.	Reject
BR-010	AITG mappings preserve ownership: an IntegrationHandoff shall not change the owner, state or decision of either framework's record.	Reject
BR-011	Inferred or automatically generated records and edges remain CANDIDATE until reviewed and shall not be inputs to an approved decision.	Flag and exclude
BR-012	Export of a record classified CONFIDENTIAL or HIGHLY_RESTRICTED requires a DistributionControl authorising the recipient and purpose.	Block export
BR-013	An Event classified as material change shall trigger impact analysis (§22) before any dependent decision is renewed.	Flag
BR-014	Risk acceptance is recorded as a decision condition or limitation; it shall not change a control result or gate result.	Reject
BR-015	Reuse of an EvidenceObject for a different assertion, scope or period requires a fresh SUPPORTS/QUALIFIES review for that assertion.	Flag

19 Canonical queries

TDS-REQ-060 Implementations claiming T4 or above shall be able to answer the queries below against a pinned snapshot, returning record identifiers and versions.

Query	Question answered
Q01	Which systems, assets and dependencies implement the functions required by Capability X?
Q02	Which failure scenarios and fault domains can defeat Outcome Y, and in which operating states?
Q03	For Control C, what evidence, results and findings exist, by claim type and scope?
Q04	For EvidenceObject E, which assertions does it support or challenge, and which decisions depend on it?
Q05	For Gate G, which criteria, CRITICAL controls, evidence and UNKNOWNs determine its result, and what ceiling follows?
Q06	For Decision D, which gates, UNKNOWNs, scores, conditions and approvals produced it?
Q07	Is every statement in Report R consistent with its decision (BR-007)?
Q08	Which records are affected by Event V (§22)?
Q09	Which providers are single points of concentration or common modes across outcomes?
Q10	What succeeds snapshot S, and by which migration?
Q11	Which candidate records await review, by age and materiality?
Q12	Which sensitive paths are visible to disclosure profile P?

20 Event model

TDS-REQ-061 Every Event shall carry: eventId, eventType (§21), source and sourceHealth, subjectRef, observedAt, effectiveAt, payload with schemaVersion, severity and materiality (candidate values until reviewed), affectedAssertions (from §22), candidateTriggers, humanDisposition (ReviewRecord), resultingDecisionEffect, the ArchitectureSnapshot in force, and supersession links where an event corrects an earlier one.

TDS-REQ-062 Events shall not change any approved record directly. An Event can only create candidate INVALIDATED_BY and TRIGGERS_REASSESSMENT edges and propose lifecycle changes, which take effect on human disposition.

21 Event types

Event type	Typical source	Typical candidate effect
ARCHITECTURE_CHANGED	Change management, design repository	New ArchitectureSnapshot required; dependent ControlAssessments flagged.
CAPACITY_THRESHOLD_BREACHED	EPMS, BMS, CDU telemetry, scheduler	CapacityAssertion INVALIDATED_BY; CG-05/CG-06 reassessment candidate.
CONFIGURATION_CHANGED	Configuration management, controller audit logs	Implementation-claim results for affected controls flagged.
PROVIDER_CHANGED	Contract or supplier management	Dependency and CG-03 reassessment candidate.
SOURCE_UNAVAILABLE	Evidence connector health	Evidence currency reduced; potential UNKNOWN.
EVIDENCE_EXPIRED	Evidence repository	EvidenceObject INVALIDATED_BY; dependent results flagged.

Event type	Typical source	Typical candidate effect
EVIDENCE_CONFLICT	Validation engine, assessor	CHALLENGES edge candidate; assertion state CONFLICTING.
INCIDENT_DETECTED	Incident management	Scenario and safeguard results flagged; decision suspension candidate.
TEST_FAILED	Test records, commissioning, exercises	Effectiveness or resilience results NONCONFORMING candidate.
GATE_STATUS_CHANGED	Gate engine	DecisionRecommendation recomputation.
FINDING_CLOSED	Action tracking	Closure review (BR-008).
CONTROL_DRIFT	Continuous control monitoring	Operation-claim results flagged.
DEPENDENCY_DRIFT	Discovery, CMDB reconciliation	Dependency records flagged; CG-03 reassessment candidate.
DECISION_SUSPENDED	Decision authority	Dependent statements and reports flagged for withdrawal or annotation.
REPORT_WITHDRAWN	Report issuer	Distribution recipients notified.

22 Change-impact contract

TDS-REQ-063 Impact analysis of an Event shall execute the steps below in order, recording each step's output in a ValidationResult linked to the Event.

1. Validate the event against its schema and source health; reject or mark UNKNOWN if invalid.
2. Resolve the subject identity and the ArchitectureSnapshot in force at effectiveAt.
3. Traverse DEPENDS_ON, COMPOSED_OF, IMPLEMENTED_BY, PROTECTED_BY, SHARES_FAULT_DOMAIN_WITH and EVIDENCED_BY edges from the subject to identify affected assertions.
4. Identify affected Controls, FailureScenarios, GateResults and ServiceOutcomes.
5. Recompute candidate coverage, scores and gate results on a copy; never on approved records.
6. Apply candidate decision ceilings from §11A and AIDCF-010.
7. Route the analysis to the record owners and the decision authority for review.
8. Record the approved disposition in a ReviewRecord.
9. Create a superseding snapshot or records where the disposition requires change.
10. Notify owners of affected statements and reports, and record the notifications.

23 API and exchange

TDS-REQ-064 An implementation exposing an exchange interface shall provide: versioned media types bound to the schema version; idempotent writes keyed by client request identifier; optimistic concurrency on record version; immutable published snapshots; pagination and filtering by class, scope, status and classification; bulk validation and dry-run modes that never persist; explicit machine-readable errors referencing the violated requirement or business rule; classification-aware access control; an append-only audit log; and no destructive overwrite of any record version.

24 Semantic operations

Operation	Behaviour
validateRecord	Validate one record against schema, shapes and business rules; return ValidationResult.
validatePackage	Validate an exchange bundle (§29) including manifest digests.
resolveIdentifier	Return the current and historical versions of an identifier and any reviewed shared identities.
getLineage	Return the inputs and provenance of a derived record.
getTraceabilityPath	Return the TR-01 to TR-18 path between two records, with gaps.
getImpactAnalysis	Execute §22 in dry-run mode.
createCandidateRecord	Create a record with reviewState CANDIDATE.
submitForReview	Route a candidate to its owner.
approveRecord	Record a ReviewRecord of type APPROVAL by an authorised person; never callable by automation alone.
supersedeRecord	Create a successor and mark the predecessor SUPERSEDED.
publishSnapshot	Freeze and digest an ArchitectureSnapshot or assessment snapshot.
exportDisclosurePackage	Export records permitted by a DistributionControl, with redaction applied.
ingestEvent	Accept an Event and start §22.
getConformanceResult	Return the latest ValidationResult set for the conformance suite.

25 Security, privacy and disclosure

TDS-REQ-065 Classification shall be enforced at record, attribute and edge level. An edge inherits at least the higher classification of its two endpoints.

TDS-REQ-066 Records and edges describing access paths to OT, BMS, EPMS, CDU, BMC/out-of-band or fabric-controller interfaces, protection settings, or security weaknesses shall be classified CONFIDENTIAL or HIGHLY_RESTRICTED and released only under a DistributionControl that names the purpose and recipients.

TDS-REQ-067 Credentials, keys, tokens and secrets shall never be stored in AIDCF records or evidence payloads; evidence of their existence or configuration shall be recorded by reference or attestation.

TDS-REQ-068 Personal data shall be minimised to what is needed for accountability (role, authority, review). Retention shall follow the AIDCF-008 evidence retention rules and applicable law.

TDS-REQ-069 Records and packages at rest and in transit shall be protected for integrity (digests and signatures per §26) and confidentiality appropriate to their classification. Queries against HIGHLY_RESTRICTED data shall be logged and monitored.

TDS-REQ-070 Content derived by AI or other automated analysis shall be labelled as such in provenance and shall remain CANDIDATE until reviewed.

TDS-REQ-071 An integrity or confidentiality breach affecting AIDCF records shall be raised as an Event and shall trigger review of all decisions whose lineage includes the affected records.

26 Provenance and integrity

TDS-REQ-072 The provenance object of every record shall carry: source system and source record; creator and collector; collection method; observation, collection and ingestion timestamps; clock source where timing matters to the assertion; transformation or migration version; SHA-256 digest; signature where the record is part of a published package; custodian and chain-of-custody entries; classification; review and disposition; SUPPORTS and CHALLENGES links; supersession and migration links. The model aligns to the W3C PROV-O concepts of entity, activity and agent.

27 Versioning and migration

Requirement	Rule
TDS-REQ-073	Schemas, ontology, registries and vocabularies shall use semantic versioning: major for breaking changes (removed field, changed meaning, tightened cardinality), minor for additive optional changes, patch for metadata and documentation.
TDS-REQ-074	Every major or minor change shall publish migration rules from the previous version, and each migration run shall produce MigrationRecords and ValidationResults.
TDS-REQ-075	Migration shall not reinterpret meaning automatically; a value without a defined mapping becomes UNKNOWN with a migration reason.
TDS-REQ-076	Deprecated terms shall remain resolvable to their replacement through `deprecated-terms.csv` and the ontology.
TDS-REQ-077	Decisions and reports whose lineage includes migrated records shall be revalidated against BR-001 to BR-015 before renewal.

28 Source-artifact binding

TDS-REQ-078 Each record shall identify its semantic owner by public ID. The binding of content to artifacts is:

Artifact	Content bound
AIDCF-002	Core concepts and the canonical assurance chain
AIDCF-003	Infrastructure ontology, taxonomy, operating states, reference classes
AIDCF-004	Threats, hazards, failure modes, scenarios, safeguards
AIDCF-005	AITG integration and IntegrationHandoff
AIDCF-006	Assurance principles
AIDCF-007A	Control domains, families, controls, gate mappings
AIDCF-008	Evidence requirements, objects, quality, sufficiency, UNKNOWN types
AIDCF-009	Assessments, procedures, samples, tests, control assessments, findings, reviews
AIDCF-010	Gates, scores, recommendations, decisions
AIDCF-011	Assurance programmes, actions, reassessment triggers, assurance levels
AIDCF-012	Statements, reports
AIDCF-013	Reference architecture, architecture snapshots, patterns and views
AIDCF-014	Manifest, identifiers, relationships, traceability, events, validation, migration
AIDCF-016	Serialisation, schema and ontology package

29 Exchange bundles

TDS-REQ-079 An exchange bundle shall contain a bundle manifest (bundle type, releaseRef, record list with digests, classification, DistributionControl), the records, and the ValidationResults produced by validatePackage. The bundle types and their minimum record sets are:

Bundle	Minimum records
Baseline	ReleaseManifest, registries, vocabularies, open BaselineIssues
Architecture	CapabilityInstance, ServiceOutcomes, RequiredFunctions, ArchitectureSnapshot and members, Dependencies, FaultDomains
Scenario	Threats, Hazards, FailureModes, FailureScenarios, Safeguards and their edges
Control	Controls in scope, EvidenceRequirements, gate mappings
Evidence	EvidencePackages and EvidenceObjects with custody and integrity
Assessment	Assessment, procedures, samples, test records, ControlAssessments, Findings, UnknownRecords
Decision	GateResults, ScoreSnapshot, DecisionRecommendation, ReviewRecords, AssuranceDecision
Reporting	AssuranceStatements, AssuranceReport, DistributionControl
Continuing	Events, ReassessmentTriggers, impact ValidationResults, Actions
Integration	IntegrationHandoffs and the referenced AIDCF records

30 Synthetic dataset

TDS-REQ-080 The AIDCF-015 synthetic dataset shall exercise every class in §7 and every relationship type in §9 at least once, and shall include: a CapabilityInstance with training and inference outcomes; a training service chain; power, cooling (including liquid cooling), compute, storage and network systems; external utility and provider dependencies; at least one shared fault domain spanning nominally redundant paths; failure scenarios with safeguards; controls with evidence; at least one UNKNOWN of each severity US2 to US5; gate results including at least one FAIL and one UNKNOWN; a score snapshot and decision; reports; a change event with reassessment; and an AITG handoff. All synthetic records shall be labelled as synthetic in provenance and shall not represent any real facility.

31 Conformance tests

TDS-REQ-081 The conformance test suite shall contain, for each test below, positive and negative fixtures, and shall be versioned and listed in the ReleaseManifest.

Test	Name	What it verifies
CT-01	Positive records	Valid fixtures for every class and relationship type are accepted.
CT-02	Missing mandatory	Records missing envelope or profile fields without nullReason are rejected.
CT-03	UNKNOWN handling	UNKNOWN results are excluded from means, counted in coverage, and severities apply the correct ceilings.
CT-04	Gate dominance	A gate FAIL or UNKNOWN blocks REASONABLY_ASSURED irrespective of scores; PARTIAL never yields PASS.
CT-05	Score reproducibility	Recomputing a ScoreSnapshot from pinned inputs yields identical values, coverage and band.
CT-06	Report consistency	Statements stronger than the decision are rejected (BR-007).
CT-07	Supersession	Superseded records remain retrievable; cycles are rejected.
CT-08	Migration	Legacy vocabulary and relationship aliases migrate correctly, with conditional mappings applied and UNKNOWN fallback.
CT-09	Security	Classified records and edges are excluded from unauthorised exports; inherited edge classification applies.
CT-10	AITG handoff	Handoffs preserve ownership and do not alter either framework's decisions.
CT-11	Event impact	A change event identifies all dependent records and changes nothing without disposition.
CT-12	Anti-gaming	Attempts to reach a stronger decision by re-scoping, reclassifying UNKNOWN as NOT_APPLICABLE, or reusing evidence across claim types are detected.
CT-13	Baseline conflict	Conflicting counts or identifiers create BLOCKED BaselineIssues.
CT-14	Performance	Validation of the reference dataset completes within the time stated in the test-suite profile (illustrative default: under 10 minutes on a single commodity server).

31A Templates

All templates carry the common envelope (§8). The fields below are the template-specific content; the envelope is not repeated.

Template	Specific fields
Release Manifest	As §4.
Baseline Issue	conflictingSources (artifact, version, location, value), conflictType, impact, status, provisionalValue, condition, expiry, resolutionRefs
Object Registry Entry	className, definition, ownerArtifact, chainStep, profileFields, schemaRef, shapeRef
Relationship Entry	typeName, definition, domain, range, cardinalityOut, cardinalityIn, inverse, symmetric, qualifiers, aliases
Traceability Link	ruleId (TR-NN), fromRef, toRef, pathEdges, result (MET, GAP, UNKNOWN), gapUnknownRef
Validation Result	subjectRef, validator and version, ruleSet, ruleId, outcome (PASS, FAIL, WARNING), message, releaseRef, runAt
Migration Record	sourceVersion, targetVersion, ruleRefs, inputRefs, outputRefs, unmappedValues, fallbackApplied, validationRefs
Impact Analysis	eventRef, snapshotRef, traversedEdges, affectedRecords, candidateEffects, reviewRef, disposition
Disclosure Profile	audience, purpose, permittedClasses, maxClassification, redactionRules, expiry, approverRef
Conformance Report	implementation, level claimed, releaseRef, testSuiteVersion, results per test, exceptions, date, reviewerRef
Integration Handoff	aidcfRefs, aitgRefs, direction, payload type, ownershipStatement, reviewRef
Source-System Mapping	sourceSystem, sourceObject, sourceField, targetClass, targetField, transformation, unit, qualityNotes
Extension Registration	As §33.
Schema Change Request	requestedChange, rationale, affectedClasses, versionImpact (major, minor, patch), migrationPlan, reviewRefs

32 Data governance roles

Role	Accountability
Methodology authority	Normative meanings, release approval.
Registry steward	Codes, registries, vocabularies and alias mappings.
Schema maintainer	AIDCF-016 schemas, context, ontology and shapes.
Domain steward	Domain-specific mappings for one of the 15 domains.
Evidence steward	Evidence semantics and quality rules.
Decision steward	Gate, score and decision logic.
Security steward	Classification and disclosure.
AITG steward	Shared identity and handoffs.
Tool implementer	Implementation fidelity and audit logging.
Independent reviewer	Challenge of conformance claims.
Release authority	Publication, manifest signature and migration.

33 Extension governance

TDS-REQ-082 An extension (class, property, relationship type or code) shall be registered before use with: a namespace prefix distinct from `aidcf:`; an owner; a justification; target classes; proof of no collision with canonical terms; datatype and cardinality; classification; JSON Schema fragment and SHACL shape; examples; tests; approval; and deprecation and migration rules. An extension shall not weaken any canonical rule (§1) and shall not be required to interpret a canonical record.

34 Data quality

TDS-REQ-083 Data quality shall be measured and reported separately from assurance confidence; good data quality does not raise confidence in a claim, and poor data quality is reported as UNKNOWN or ValidationResult failures.

Dimension	Measure
Completeness	Mandatory fields and traceability links present (§8, §11).
Validity	Values conform to schema and vocabularies.
Consistency	No contradictions between records, versions or sources.
Uniqueness	No duplicate identities.
Timeliness	Records current relative to their effectiveTo and source update frequency.

Dimension	Measure
Accuracy	Reconciliation to source systems and observed state.
Lineage	Provenance complete (§26).
Integrity	Digests and signatures verify.
Reviewability	Proportion of material records with ReviewRecords.
Security	Classification and DistributionControl coverage.

35 Reference implementation components

Component	Responsibility
ArtifactRepository	Stores artifacts and their digests.
RegistryRepository	Stores registries and vocabularies by version.
SchemaRegistry	Serves the AIDCF-016 package by version.
IdentifierService	Mints, resolves and checks collisions of identifiers.
IngestionService	Imports source-system data as candidate records.
ValidationEngine	Runs schema, shape, traceability and business-rule validation.
TraceabilityGraph	Holds records and edges with versions and named snapshots.
EvidenceConnectors	Collect evidence with provenance and integrity.
AssessmentWorkflow	Supports procedures, sampling, tests and review.
GateScoreEngine	Computes gate results and scores reproducibly.
DecisionReviewWorkflow	Routes recommendations to authority and records approvals.
ReportingService	Generates statements and reports and checks BR-007.
EventImpactService	Ingests events and runs §22.
DisclosureService	Applies DistributionControls to exports.
AuditLog	Append-only log of all writes and approvals.
ConformanceRunner	Runs the §31 suite and publishes ValidationResults.

These are logical components; AIDCF-015 provides a reference implementation.

36 Storage projections

TDS-REQ-084 Any storage projection (graph database, triple store, relational, search index, warehouse, cache) shall: retain the canonical record or a lossless reference to it; preserve edge provenance and version; use named graphs or equivalent for snapshots and inferred data; use stable keys equal to AIDCF identifiers; treat search indexes and caches as non-authoritative; reference evidence blobs by digest rather than copy them into derived stores; and never approve or overwrite canonical records.

36A Worked traces

The traces below are illustrative and synthetic.

36A.1 Electrical

Outcome "training job availability within tolerance" REQUIRES_FUNCTION "continuous power to accelerator racks"; IMPLEMENTED_BY the rack power train (UPS system, distribution, busway, rack PDUs). The UPS DEPENDS_ON utility and generator (Dependency strength MATERIAL in NORMAL and EMERGENCY). Scenario: a utility voltage sag INITIATES transfer to UPS; PROPAGATES_TO generator start and load step. Safeguards: UPS ride-through, generator load acceptance, load-step management by the cluster power-capping function. Controls MCF-POW-011 (Runtime and Derating) and MCF-POW-020 (Load-Step Response) are ASSESSED_BY ControlAssessments; their ControlAssessments are linked by EVALUATES_GATE to the CG-05 GateResult. If generator load-acceptance test evidence is older than its currency floor, the result is UNKNOWN, HAS_UNKNOWN a US4 record, and CG-05 is UNKNOWN.

36A.2 Thermal

Outcome REQUIRES_FUNCTION "remove rack heat at design load". It is IMPLEMENTED_BY the CDU secondary loop and the facility water system. Two nominally redundant CDUs SHARES_FAULT_DOMAIN_WITH each other through a common BMS controller (CONTROLLED_BY). Scenario: controller failure stops both CDU pumps; PROPAGATES_TO accelerator thermal throttling and job failure. Controls MCF-THM-010 (CDU Capacity) and MCF-THM-021 (Control Stability) are linked to CG-06, CG-07 and CG-09. Independence evidence that is absent yields a QUALIFIES edge or an UnknownRecord on the redundancy claim.

36A.3 Digital authority

Outcome depends on cooling and power control. The BMS, EPMS and BMC/out-of-band management network are CONTROLLED_BY PartyEntities acting through privileged identities. A remote-support path (SUP-06) CONNECTS_TO the OT gateway. Controls MCF-SEC-005 (BMS and EPMS Security) and MCF-NET-016 (Break-Glass Access) link to CG-07 and CG-15. The access-path records are HIGHLY_RESTRICTED (§25).

36A.4 Recovery

Outcome "resume training from checkpoint within recovery tolerance" RECOVERS_THROUGH the checkpoint tier and alternate capacity. The checkpoint tier DEPENDS_ON the storage fabric and the scheduler. Controls MCF-STO-017 (Checkpoint Durability), MCF-STO-019 (Storage Recovery Sequencing) and MCF-RES-016 (Data and Checkpoint Recovery) link to CG-10; the restore exercise TestRecord RESULTS_IN their effectiveness assessments.

36A.5 Report lineage

AssuranceDecision (CONDITIONALLY_ASSURED, ACTIVE) is RECOMMENDS-linked from its DecisionRecommendation, CONSTRAINS-linked from a US3 UnknownRecord (condition), APPROVED_BY the authority's ReviewRecord, and REPORTED_IN an AssuranceReport. An EVIDENCE_EXPIRED event later INVALIDATED_BY-links a decisive EvidenceObject; on disposition the decision lifecycle moves to SUSPENDED and the report owner is notified.

37 Conformance validation

Check	Verification	Requirements verified
TDS-VAL-001	ReleaseManifest present, complete, signed and digests verify.	TDS-REQ-022, TDS-REQ-023
TDS-VAL-002	All source conflicts recorded as BaselineIssues.	TDS-REQ-003, TDS-REQ-035
TDS-VAL-003	Identifiers stable, unique and pattern-conformant.	TDS-REQ-027, TDS-REQ-030
TDS-VAL-004	Envelope fields present (owner, source, version, status).	TDS-REQ-037
TDS-VAL-005	Edges conform to domain, range and cardinality, or gaps are UNKNOWN.	TDS-REQ-039, TDS-REQ-042
TDS-VAL-006	NOT_APPLICABLE justified and approved.	TDS-REQ-059
TDS-VAL-007	Material assertions evidenced or UNKNOWN.	TDS-REQ-010
TDS-VAL-008	Evidence quality evaluated per claim type.	TDS-REQ-013
TDS-VAL-009	Results link method, subject, evidence and reviewer.	TDS-REQ-042
TDS-VAL-010	Gate PASS has evidence at floor and conforming CRITICAL controls.	TDS-REQ-059
TDS-VAL-011	Scores reproducible.	TDS-REQ-012
TDS-VAL-012	Decision respects gates and UNKNOWN ceilings.	TDS-REQ-059
TDS-VAL-013	Reports do not exceed decisions.	TDS-REQ-059
TDS-VAL-014	Events do not auto-approve.	TDS-REQ-062, TDS-REQ-006
TDS-VAL-015	History preserved.	TDS-REQ-007
TDS-VAL-016	Migrations explicit and recorded.	TDS-REQ-044, TDS-REQ-040, TDS-REQ-074
TDS-VAL-017	AITG ownership preserved.	TDS-REQ-018
TDS-VAL-018	Sensitive paths classified and controlled.	TDS-REQ-020, TDS-REQ-066
TDS-VAL-019	Extensions registered and non-colliding.	TDS-REQ-082
TDS-VAL-020	Conformance tests pass for the claimed level.	TDS-REQ-021, TDS-REQ-081

38 Review and release gates

All reviews below are pending. No review, pilot or independent conformance assessment of this standard has been completed.

Reviewer or pilot	Status
Methodology author	Pending
Ontology and data architect	Pending
Control and evidence reviewer	Pending
Assessment reviewer	Pending
Scoring and decision reviewer	Pending
Programme and reporting reviewer	Pending
Reference-architecture reviewer	Pending
JSON Schema reviewer	Pending

Reviewer or pilot	Status
RDF, OWL and SHACL reviewer	Pending
Security and privacy reviewer	Pending
AITG reviewer	Pending
Synthetic dataset pilot	Pending
Graph pilot	Pending
API and event pilot	Pending
Independent conformance review	Pending
Public release	Pending

Final doctrine

AIDCF-014 makes assurance claims reconstructable and testable without giving machines authority they do not possess. Structural validity does not prove truth, engineering adequacy or assurance. Conflict, incomplete lineage and missing evidence remain explicit and prevent stronger conclusions.